Féidearthachtaí as Cuimse
Infinite Possibilities

Secure Communications & Cryptography (MACS H6014)

Classic Ciphers

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Getting Started

- **What you should complete during this week:**

1. Enroll on the Brightspace module page

2. Join the class forum on Discord
   - Post and introduce yourself

3. Create an account on Cryptohack.org
   - Attempt the 3 challenges from the introduction section

4. Get a copy of the Course textbook
   - Read Chapter 1: Ciphers and Fundamentals

muhammad.arshad@tudublin.ie

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Brief Timeline

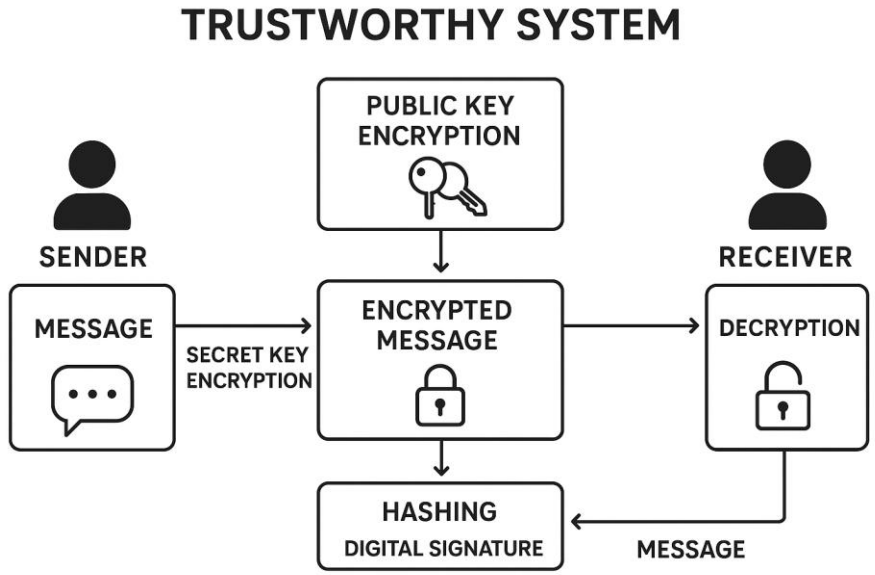
- Substitution ciphers e.g. Caesar / Vigenère
- One-Time Pad
- Enigma Machine
- Many people regard the *"Communication Theory and Secrecy Systems"* paper, by **Claude Shannon** in 1949, as the foundation of modern information security theory.
- In 1976, the paper "New Directions in Cryptography," by Whitfield Diffie and Martin Hellman, (and RSA in 1978) showed that communications over an insecure line can be secure with no need for a common secret key.
- Public Key Exchange: solution to secure information exchange
- Fully Homomorphic Encryption (FHE)

muhammad.arshad@tudublin.ie

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What is the Problem

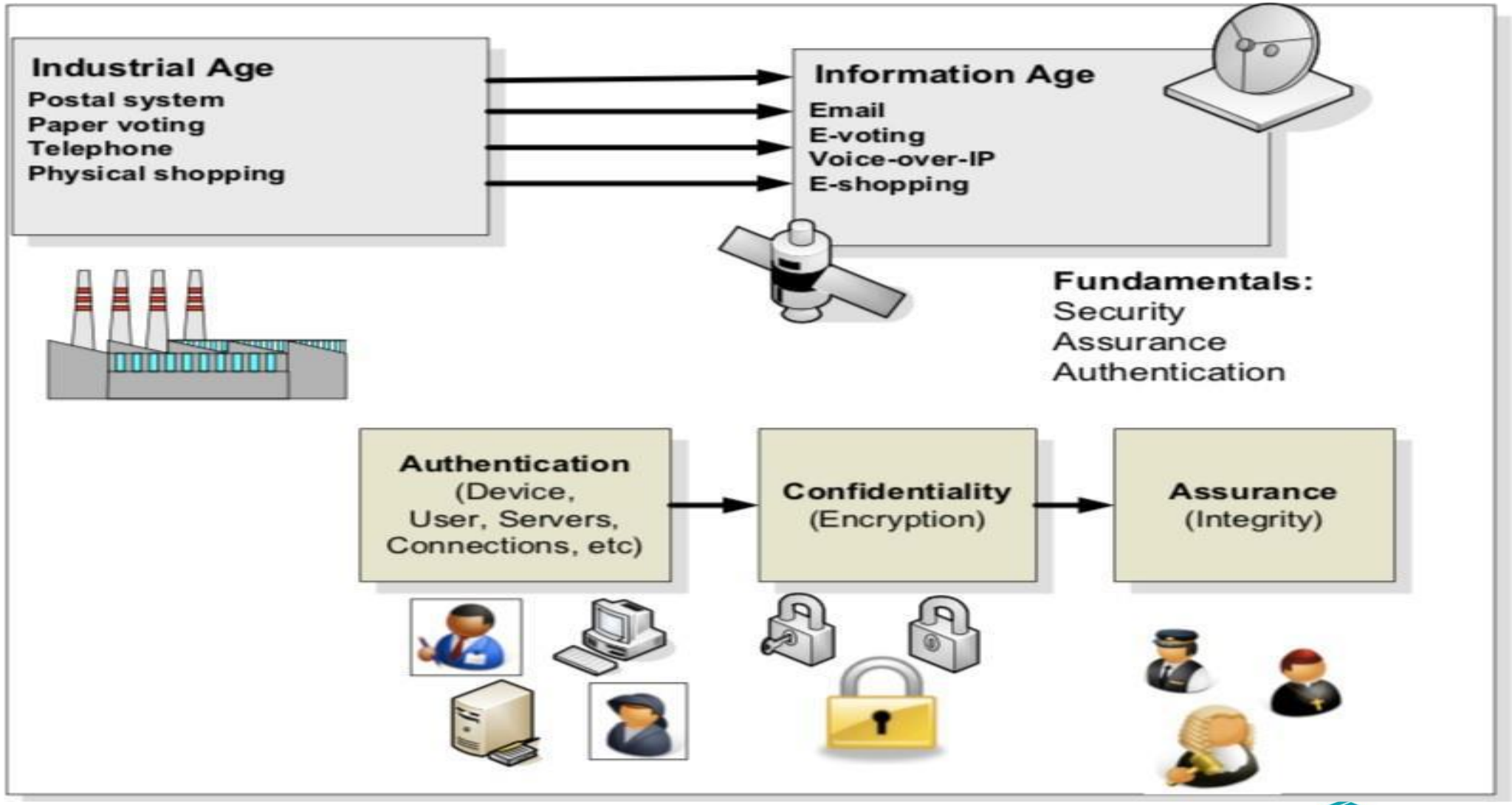
- The Internet is built on services and protocols that **lack security** at the core.
- We **can't be sure** that a message we receive is actually from the sender or that the contents of the message have not been seen (or altered) by a third party.
- What is **required is a trustworthy system,** and this requires systems based on cryptography to ensure secrecy and validation.
- Unfortunately, **no one technique** will provide this, and today we **use a mixture of methods to create trustworthiness**.
- For example, we use **secret key encryption** to protect the contents of messages and **public key encryption** to prove the identity of the sender and receiver.
- We use **hashing methods** to provide the validity of messages and **digital signatures** to prove the identity of the sender and the contents of the message.



muhammad.arshad@tudublin.ie

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Real World Scenario



muhammad.arshad@tudublin.ie

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Secure Network

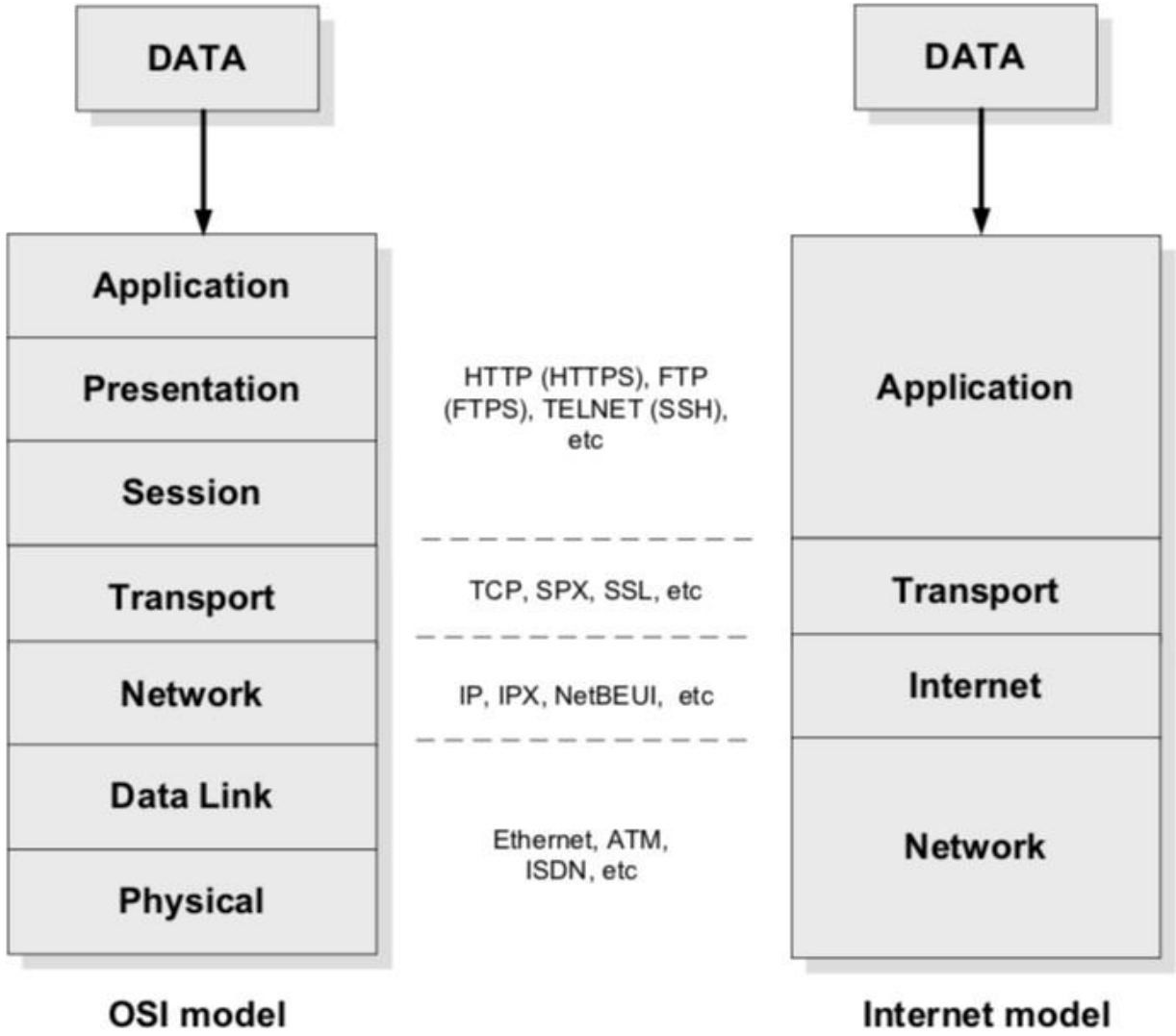
- Network infrastructure devices and hosts are secured using a variety of techniques:
  - Device hardening
  - AAA (Authorization, and Accounting) access control
  - Access Control Lists (ACLs)
  - Firewalls
  - Monitoring threats using an **intrusion detection system (IDS)** and/or **intrusion prevention system (IPS)**.
  - Securing endpoints using Advanced Malware Protection (AMP)
  - Enforcing email and web security

- The next goal is to **secure the data** as it travels across various links. This may include internal traffic, but of greater concern is protecting the data that travels outside of the organization to branch sites, telecommuter sites, and partner sites, especially **across public networks**.

muhammad.arshad@tudublin.ie

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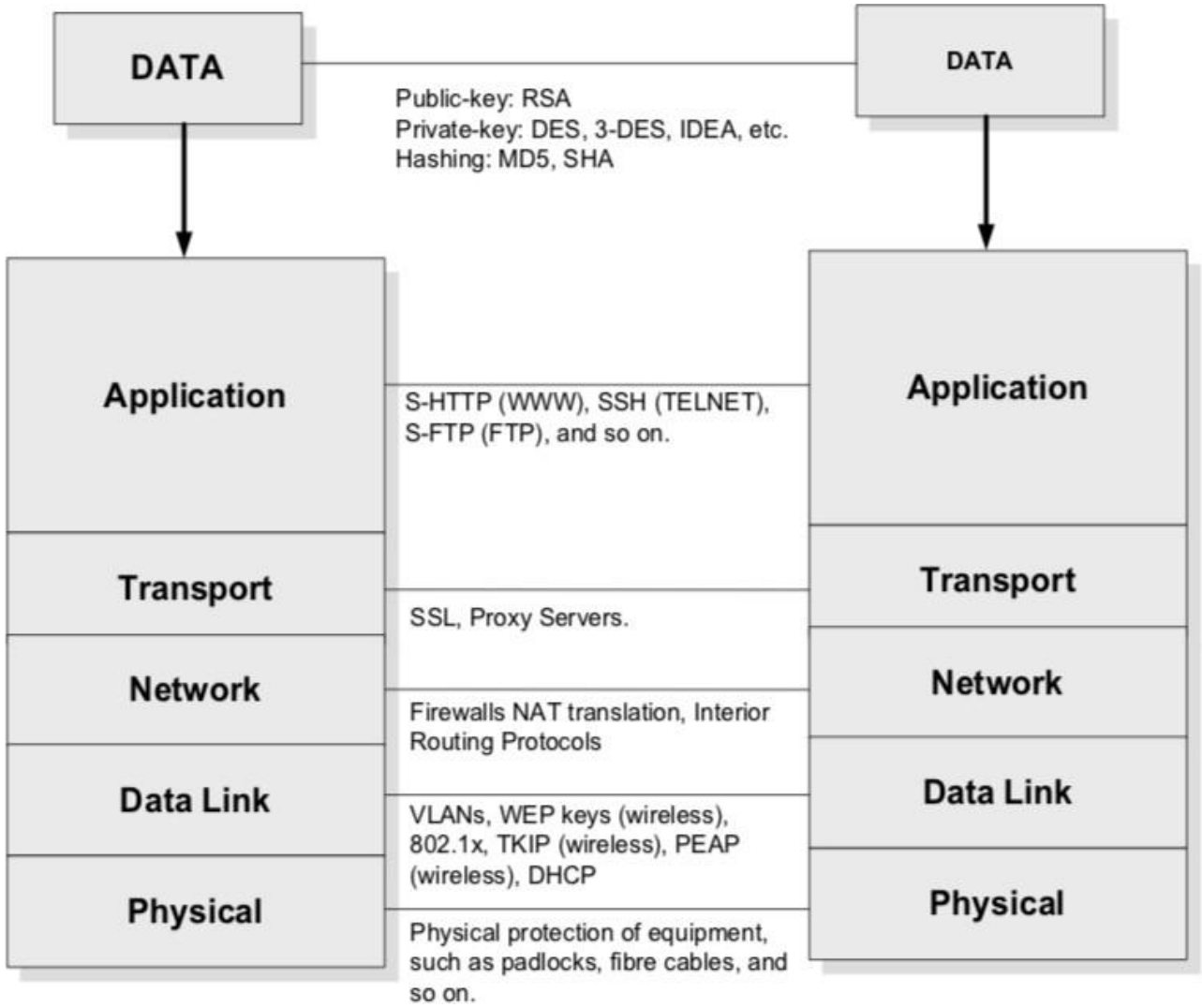
Secure Network (Contd.)



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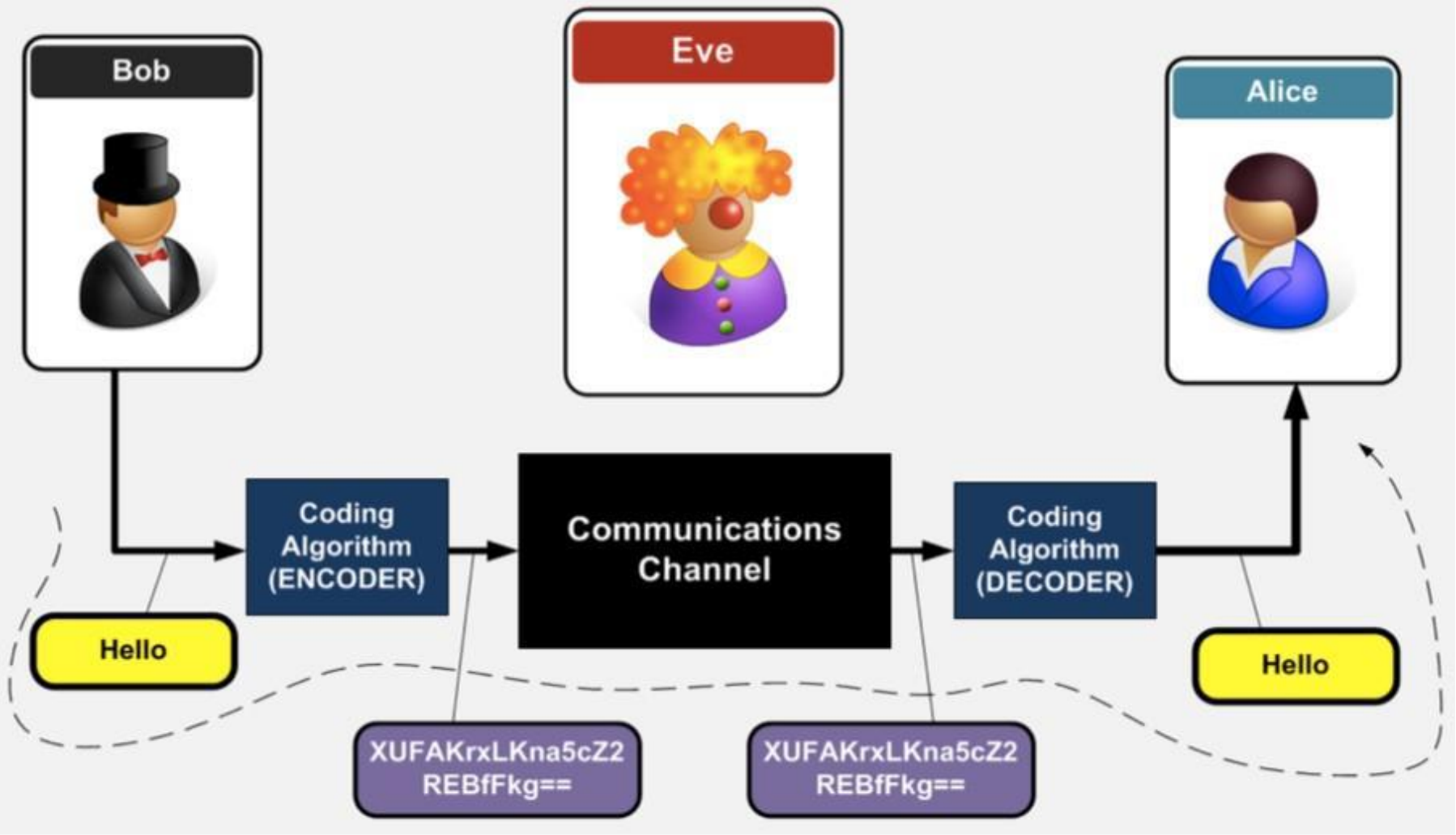
Secure Network (Contd.)



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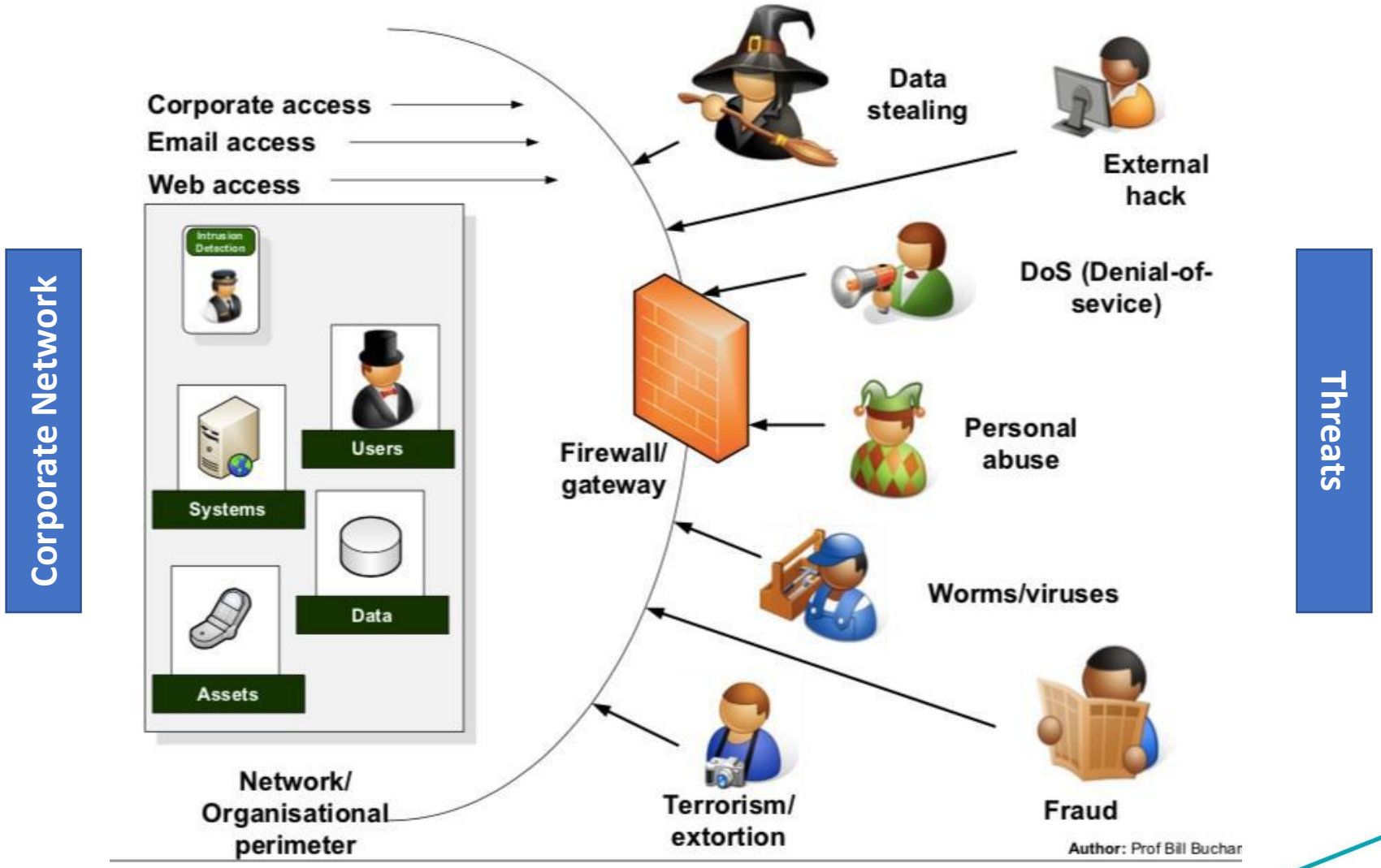
Basic Idea Behind Secure Communication



muhammad.arshad@tudublin.ie

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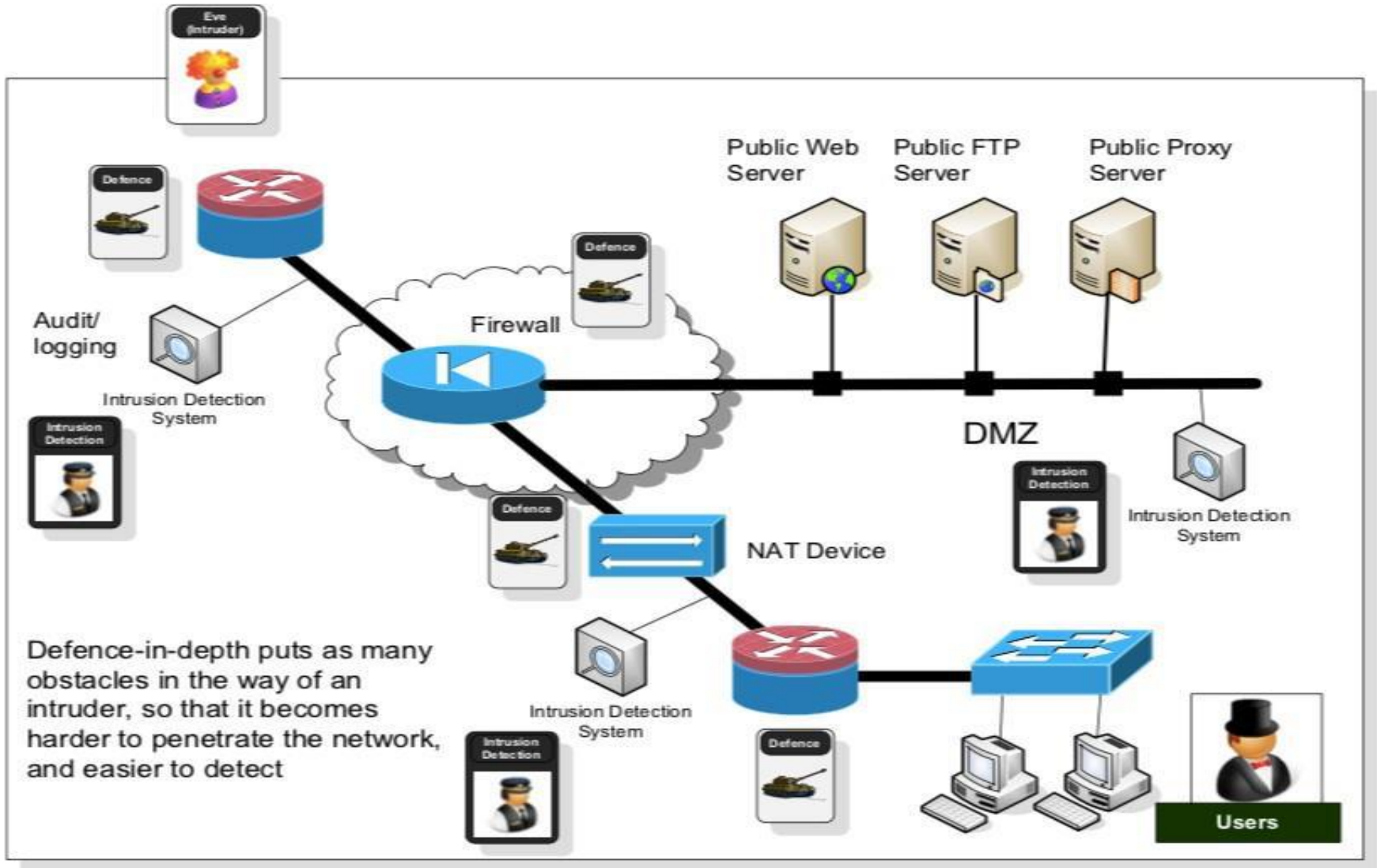
Corporate Networks and Threats



muhammad.arshad@tudublin.ie

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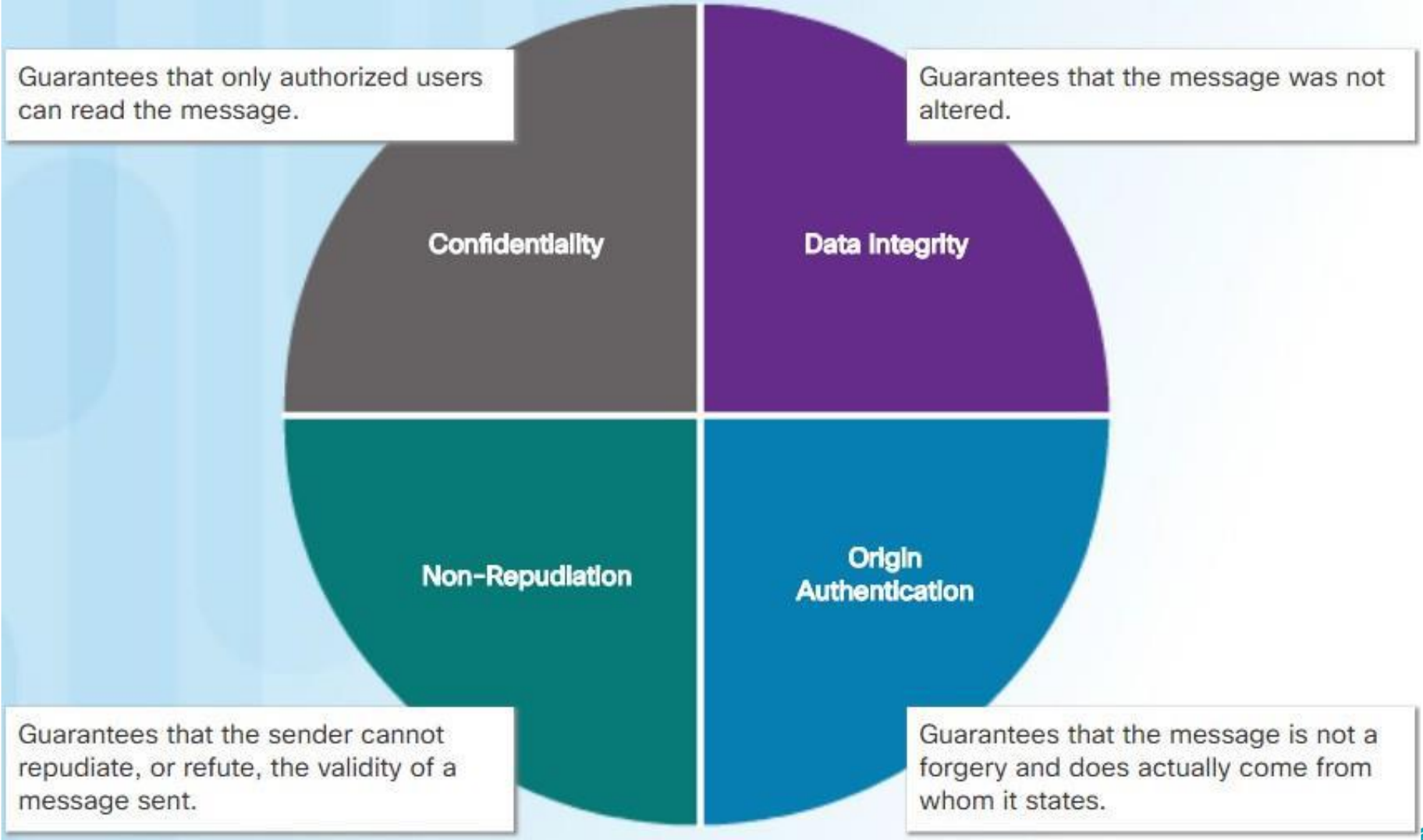
Enterprise Network Architecture



muhammad.arshad@tudublin.ie

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Secure Communications Elements



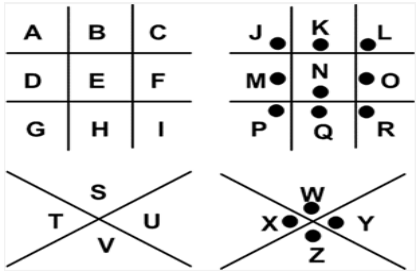
muhammad.arshad@tudublin.ie

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Secure Communications

Since the beginning of **human communication**, we have attempted to conduct it secretly.









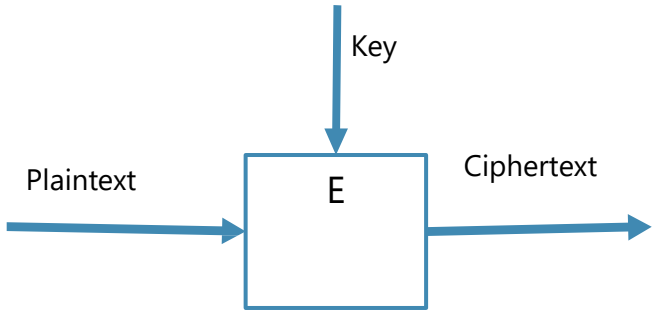
Plaintext	h	e	l	l	o	e	v	e	r	y	o	n	e
Morse:	....	.	.-..	.-..	---	.	...-	.	.-.	-.--	---	-.	.

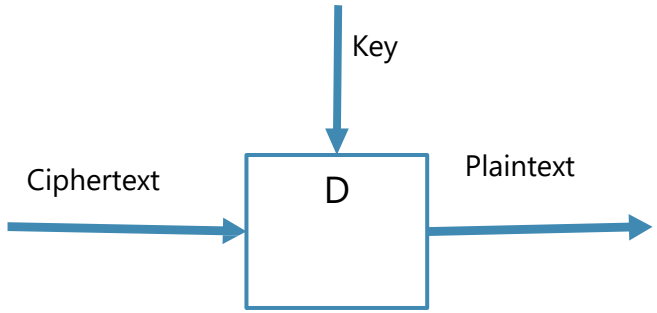
muhammad.arshad@tudublin.ie

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What is Cipher

A pair of functions (Encryption & Decryption) that satisfy the consistency equation.





C = E(k,P) **P = D(k,C)**

P = D(k, E(k, P)) : Consistency Equation

muhammad.arshad@tudublin.ie

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Types of Classical Ciphers (Pen and Paper)

1. Substitution Ciphers
2. Transposition Ciphers
3. Block Ciphers
4. Stream Ciphers
5. Polyalphabetic Ciphers
6. Public-Key Cryptography
7. One-Time Pad

muhammad.arshad@tudublin.ie

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Substitution Ciphers

- Monoalphabetic Substitutions (Simple 1 to 1 mapping)
  - *Caeser Cipher*

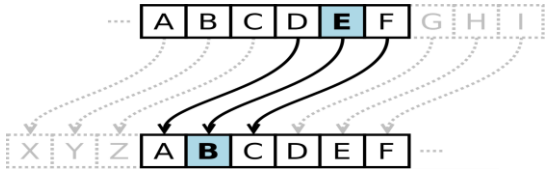
    Example: With a shift of 3 positions (Caesar cipher with a key of 3):
    Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
    Cipher: X Y Z A B C D E F G H I J K L M N O P Q R S T U V W
  - *Atbash Cipher*

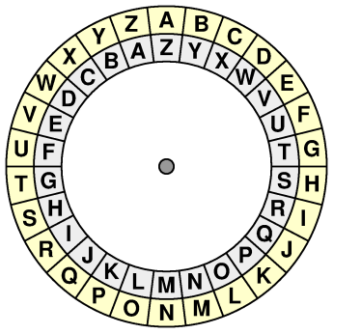
    Example: (reverse in the alphabet)
    Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
    Cipher: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
  - *Simple Substitution Cipher*

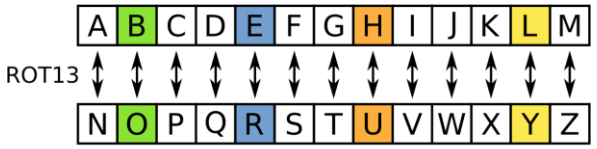
    Example (random mapping):
    Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
    Cipher: P Q R S T U V W X Y Z A B C D E F G H I J K L M N O
  - *Keyword Cipher*

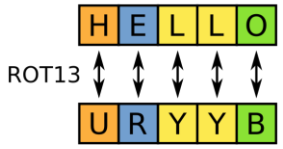
    Example (Keyword: KEYWORD):
    Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
    Cipher: K E Y W O R D A B C F G H I J L M N P Q S T U V X Z
  - *ROT13 (Rotate by 13 places)*

    Example (with 13 places)
    Plain: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
    Cipher: N O P Q R S T U V W X Y Z A B C D E F G H I J K L M









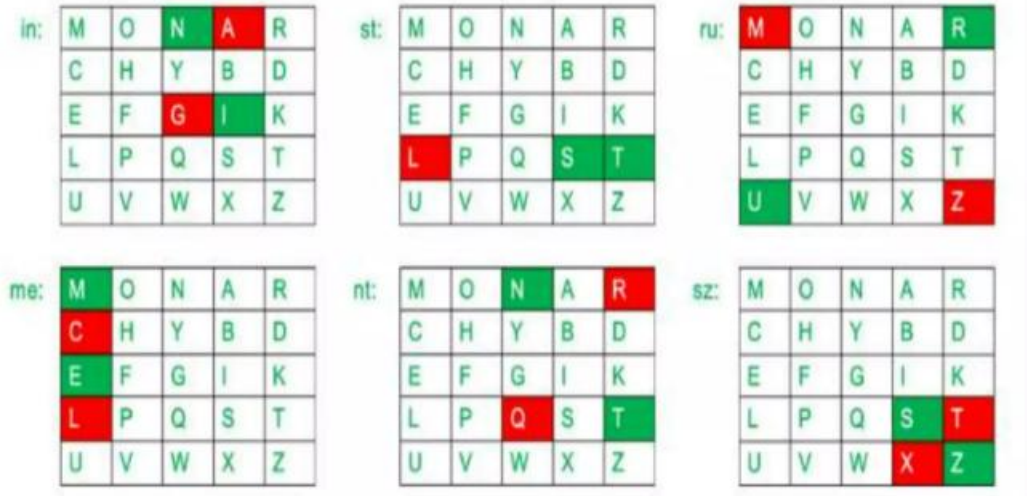
muhammad.arshad@tudublin.ie

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Substitution Ciphers (Contd.)

- Polyalphabetic Substitutions (Simple 1-to-many mapping usually based on position)
  - *Vigenère Cipher* ***(Ci=(Pi+Ki)mod26), where: Pi= position of plaintext letter (A=0, B=1, … Z=25), Ki = position of keyword letter, and Ci = ciphertext letter***

    Example (Keyword: KEY):
    Plain: H E L L O W O R L D
    Keyword: K E Y K E Y K E Y K
    Cipher: R I J V S U Y V J H
- Polygraphic Substitution (Simple 1 to 1 mapping, but a group of letters)
  - Same row → replace with letter to the right.
  - Same column → replace with letter below.
  - Rectangle → replace with letters at the corners of the rectangle.




```
Plain Text: "instrumentsz"
Encrypted Text: gatlmzclrqtx
Encryption:
  i -> g
  n -> a
  s -> t
  t -> l
  r -> m
  u -> z
  m -> c
  e -> l
  n -> r
  t -> q
  s -> t
  z -> x
```

muhammad.arshad@tudublin.ie

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Substitution Ciphers (Contd.)

- Rotational / Shift Ciphers

Caesar Cipher (Shift 1):

- Original: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
- Shifted: B C D E F G H I J K L M N O P Q R S T U V W X Y Z A

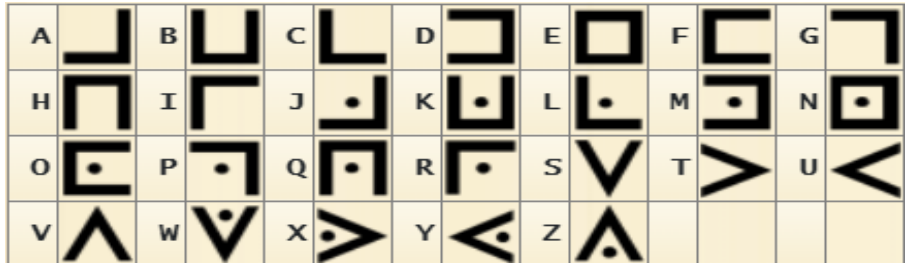
Caesar Cipher (Shift 3):

- Original: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
- Shifted: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Caesar Cipher (Shift 13, also known as ROT13):

- Original: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
- Shifted: N O P Q R S T U V W X Y Z A B C D E F G H I J K L M

- Pigpen Cipher (Masonic Cipher)







muhammad.arshad@tudublin.ie

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Substitution Ciphers (Contd.)

- Homonphonic Ciphers
  - Plaintext is replaced by multiple characters (symbols or digits) in the ciphertext
  - The letter 'E' might be replaced by any of 5 different symbols, while the letter 'Q' may only be substituted by 1 symbol

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	X	S	F	Z	E	H	C	V	I	T	P	G	A	Q	L	K	J	R	U	O	W	M	Y	B	N
9				7				3					5	0				4	6						
				2																					
				1																					

```
plaintext:  DEFEND THE EAST WALL OF THE CASTLE
ciphertext: F7EZ5F UC2 1DR6 M9PP 0E 6CZ SD4UP1
```

muhammad.arshad@tudublin.ie

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Transpositional Ciphers

- Scytale Cipher: (Rows 4) Plaintext message: **MEETATDAWN** – Ciphertext **MAWETNEDXTAX**,
  (Rows 5) Plaintext message: **THEATTACKISATDAWN** – Ciphertext **TTSWHAANECTXAKDXTIAX**

Write across in rows of 4:

1. Row 1: `M E E T`
2. Row 2: `A T D A`
3. Row 3: `W N X X` *(with padding if needed)*

Row 1: `T H E A T`

Row 2: `T A C K I`

Row 3: `S A T D A`

Row 4: `W N X X X` *(padded to fill the row)*

Column 1: `T T S W`

Column 2: `H A A N`

Column 3: `E C T X`

Column 4: `A K D X`

Column 5: `T I A X`

- Rail Fence Cipher (3 Rails)

```
W . . . E . . . C . . . R . . . U . . . O . . .
. E . R . D . S . O . E . E . R . N . T . N . E
. . A . . . I . . . V . . . D . . . A . . . C .
```

```
WECRUO ERDSOEERNTNE AIVDAC
```

```
W   E   C   R   U   O
 E R D S O E E R N T N E
  A   I   V   D   A   C
```

muhammad.arshad@tudublin.ie

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Transposition Ciphers

- Columnar Transposition Cipher
  - Characters of the plaintext are rearranged according to a specific columnar arrangement
  - These ciphers are more complex and provide better security
  - The plaintext is divided into groups of adjacent letters

Encryption

Given text = Geeks for Geeks

Keyword = HACK **Length of Keyword** = 4 (no of rows) **Order of Alphabets in HACK** = 3124

H	A	C	K
3	1	2	4
G	e	e	k
s	_	f	o
r	_	G	e
e	k	s	_

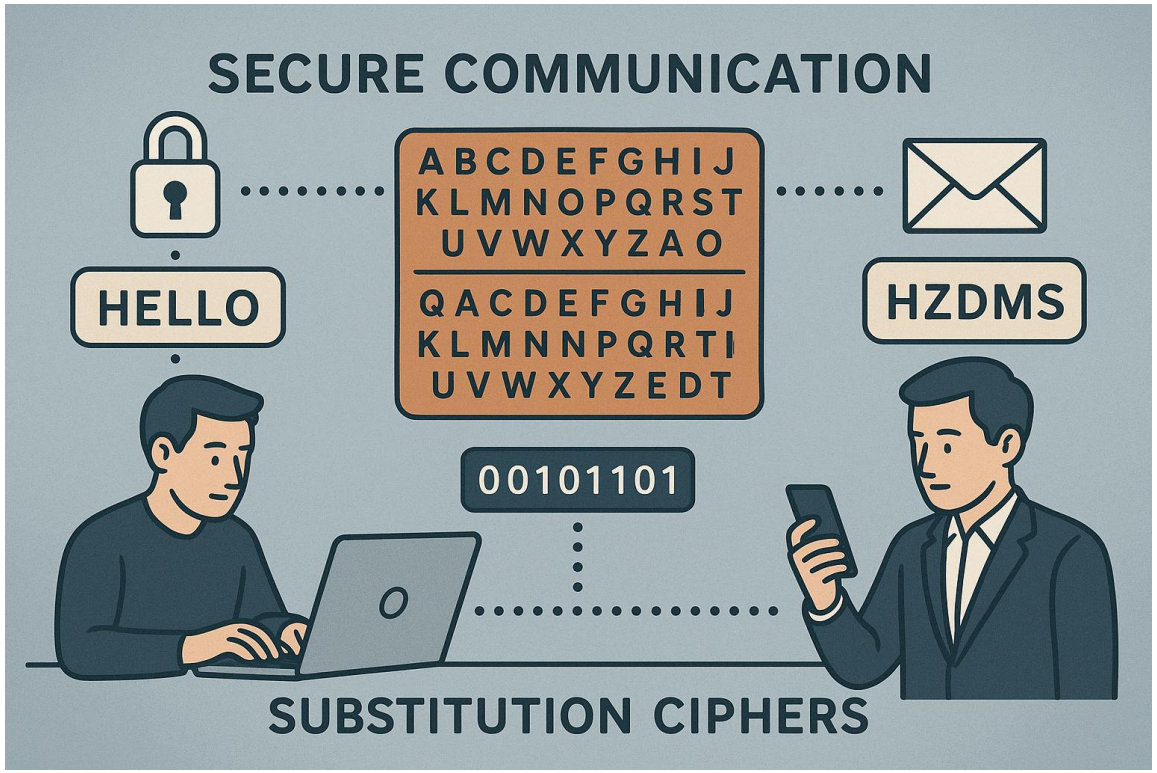
Print Characters of column 1,2,3,4

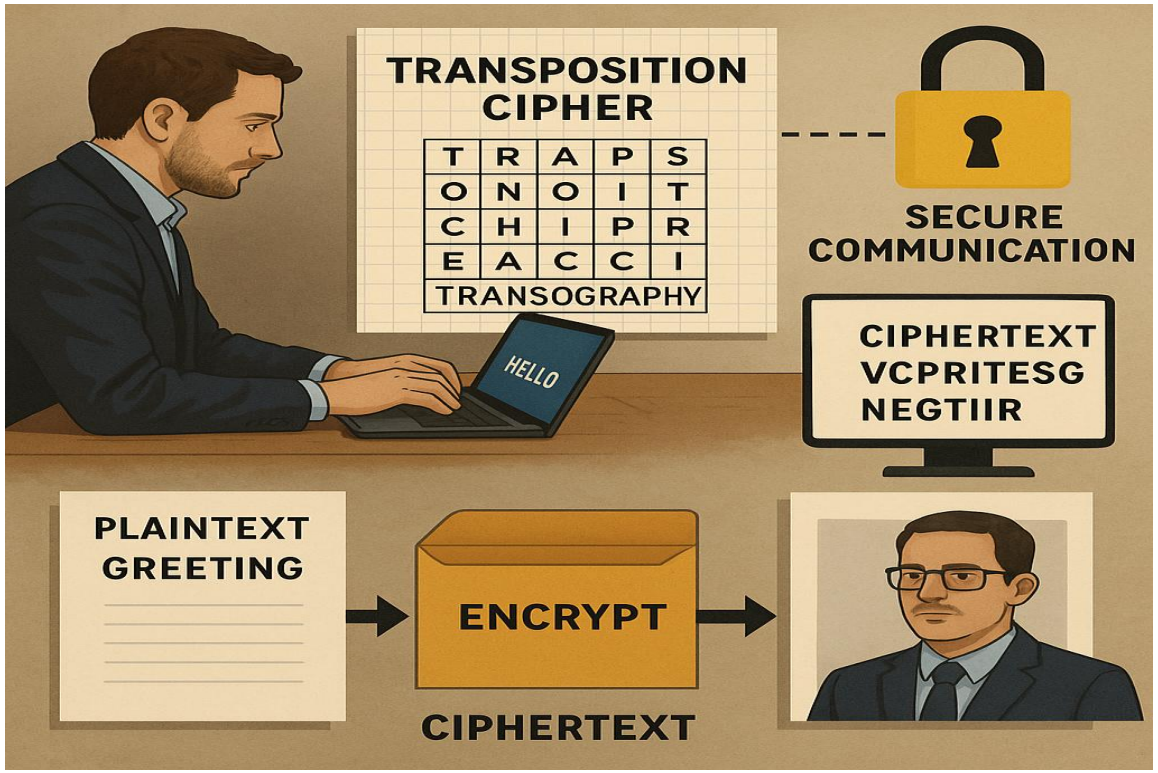
Encrypted Text = e kefGsGsrekoe_

muhammad.arshad@tudublin.ie

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Real World Implications of Ciphers

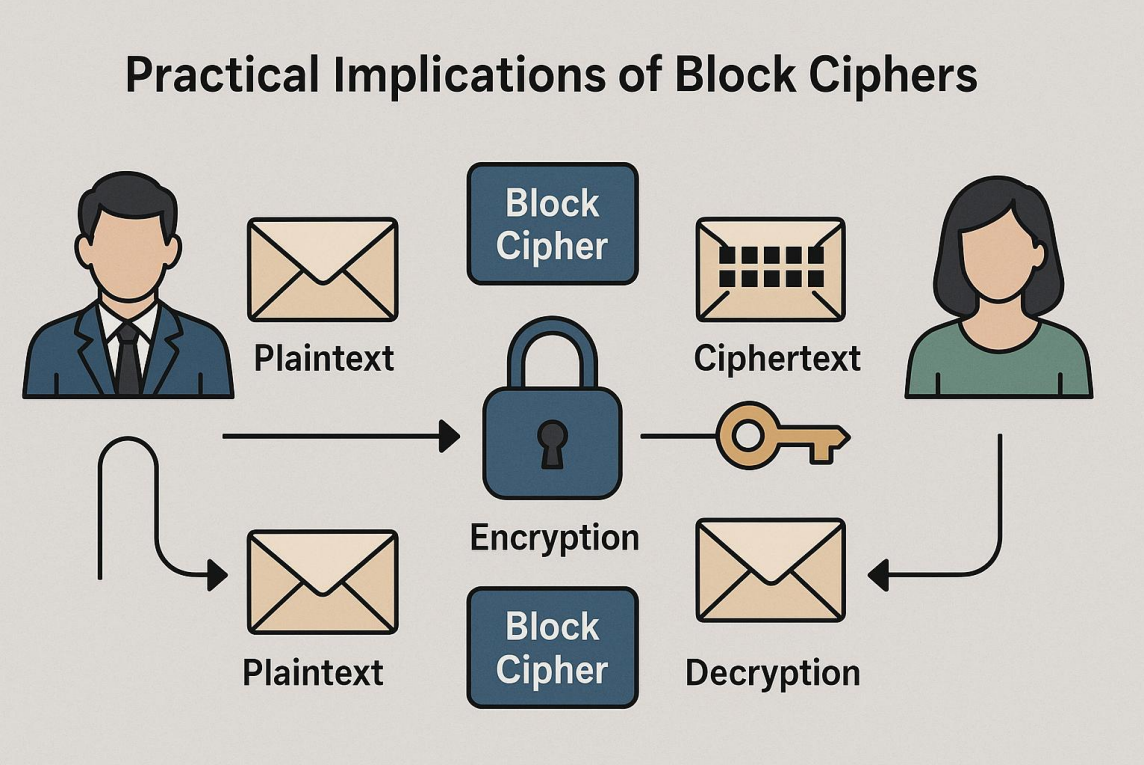


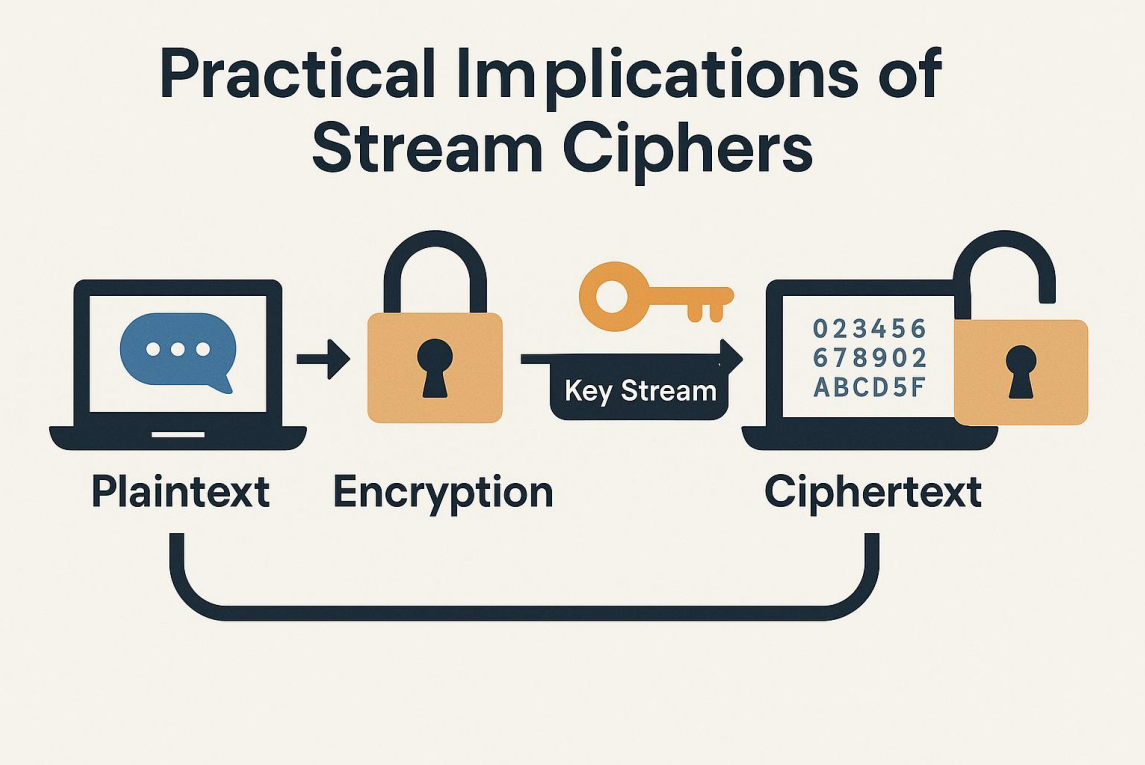


muhammad.arshad@tudublin.ie

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Real World Implications of Ciphers (Contd.)

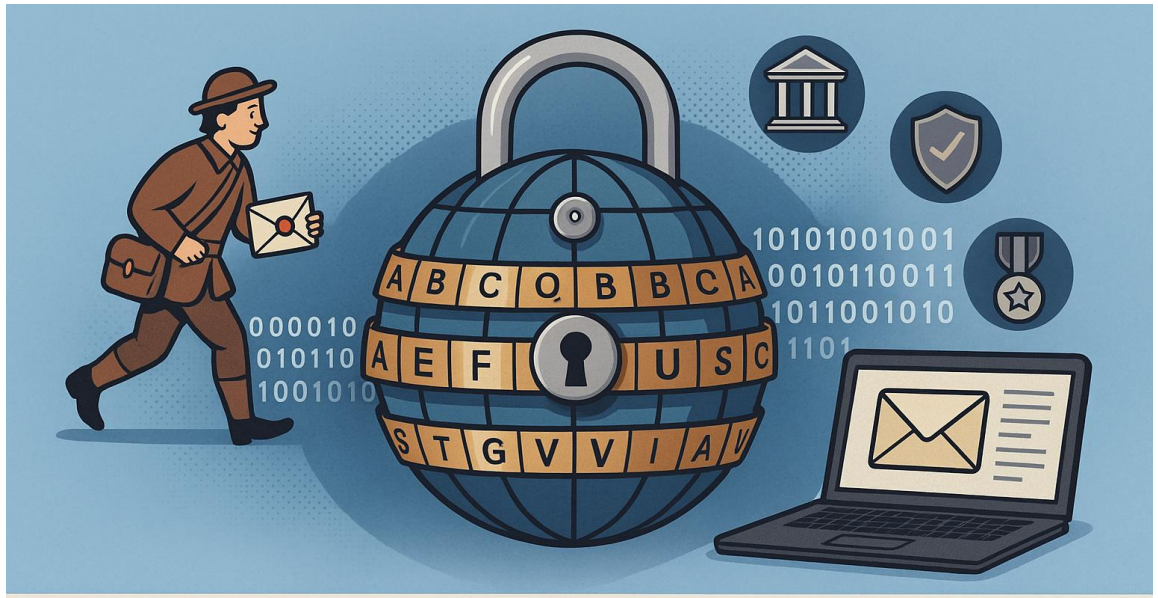




muhammad.arshad@tudublin.ie

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Real World Implications of Ciphers (Contd.)



Polyalphabetic ciphers, though historical, laid the foundation for modern cryptography by introducing complexity, reducing frequency analysis attacks, and shaping today's secure digital comniu-nication systems.



Public-key cryptography enables secure digital communi-cation by providing a scalable method for encrypting and decrypting sensitive information using pairs of cryptographic keys.

muhammad.arshad@tudublin.ie

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Fruitful Websites

- **Cyberchef.com**
  - CyberChef, the creator is GCHQ. Well, it's a government intelligence agency, in fact, it's the UK's Government Communications Headquarters.
  - You can use CyberChef to compress or decompress data using algorithms like Gzip, Bzip2, and Zip.
  - CyberChef includes various cryptographic operations like AES encryption, RSA encryption, and hashing algorithms such as MD5, SHA-1, and SHA-256.

- **dcode.fr**
  - A website that provides various online tools and resources related to cryptography, ciphers, codes, and puzzles.
  - It offers a wide range of tools and services that can help users decode, encode, or solve different types of codes and ciphers.
  - Cipher solvers, Cryptanalysis, Puzzle solvers, Encoding and decoding tools, and Cryptography information, etc.

- **asecuritysite.com**
  - The website offers a wide range of educational resources and tools related to various aspects of computer and information security.
  - It serves as a valuable resource for both beginners and experts in the field of cybersecurity.
  - Security challenges, Online tools, Security news, Security research, and Educational materials, etc.

muhammad.arshad@tudublin.ie

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Short Quiz

- **Encrypt and Decrypt the HELLO WORLD by Ceaser cipher?**
  - The solution is “KHOOR ZRUOG”

- **Encrypt and Decrypt the HELLO WORLD by ROT13?**
  - The solution is “URYYB JBEYQ”

- **Encrypt and Decrypt by Columnar Transposition Cipher, plain text is THIS IS WIKIPEDIA and the key is CIPHER.**
  - The solution is

```
C I P H E R     Key
1 4 5 3 2 6     Sequence (key letters in alphabetical order)
T H I S I S     Plaintext
W I K I P E
D I A * * *

Ciphertext by column:
  #1 TWD, #2 IP, #3 SI, #4 HII, #5 IKA, #6 SE
Ciphertext in groups of 5 for readability:
    TWDIP*SI*HIIIKASEE
```

muhammad.arshad@tudublin.ie

Practical Exercise

- **Week 2 - 25th Sept**
  - Complete Lab 1 - Setting up your **Python/ virtual lab** environment
  - Complete **Lab 1A - Ciphers and Fundamentals and Lab 1B - Numbers Station** Worksheets
  - Lab 1C – **Many Time Pad (Challenge Lab)** due on **4th October**.
  - Read Chapter 1 – **Ciphers and Fundamentals** – Cryptography by William Buchanan
  - Register for the **CryptoHack.org** website – Complete the Introduction challenges
  - Join the class **Discord** server
  - **Test 1** on **23rd October** during the lab session

muhammad.arshad@tudublin.ie